MIL-B-44314
30 DEC 1987
SUPERSEDING
(see 6.11)

MILITARY SPECIFICATION

BODY ARMOR, SMALL ARMS PROTECTIVE, AIRCREWMAN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers body armor to be worn by aircrewman for protection against small arms projectiles.

1.2 Classification. The body armor shall be of the following types, classes, and sizes (see 6.2).

Type I	-	Front insert with vest
Type II	-	Front insert and back insert with vest
Type III	-	Front insert without vest
Type IV	-	Back insert without vest
Class 1	-	Areal density of 9.0 lbs. per sq. ft., without spall cover, (aluminum oxide type)
Class 2	-	Areal density of 8.0 lbs. per sq. ft., without spall cover, (silicon carbide type)
Class 3	-	Areal density of 7.5 lbs. per sq. ft., without spall cover, (boron carbide/silicon carbide/silicon type or boron carbide/silicon type)
Class 4	-	Areal density of 6.9 lbs. per sq. ft., without spall cover, (boron carbide type)
Size	-	Short
Size	-	Regular
Size	-	Long

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8470

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

CCC-P-91	-	Padding; Rubberized, Curled Hair
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulostic
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-C-12369	-	Cloth, Ballistic, Nylon
MIL-V-43544	-	Vest, Small Arms Protective Body Armor

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-453	-	Inspection, Radiographic

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-2-217	-	Insert, Small Arms Protective Body Armor, Aircrewman, Front
8-2-218	-	Insert, Small Arms Protective Body Armor, Aircrewman, Back

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 635	-	Test Methods for Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position
D 792	-	Methods of Test for Specific Gravity and Density of Plastics by Displacement
D 2000	-	Rubber Products in Automotive Applications
D 2584	-	Ignition Loss of Cured Reinforced Resins
D 3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials (see 6.10).

3.2.1 Ceramic. The ceramic material for each of the classes of Body armor shall be as follows:

Class 1	-	Aluminum oxide
Class 2	-	Silicon carbide
Class 3	-	Boron carbide/silicon carbide/silicon or boron carbide/silicon
Class 4	-	Boron carbide

3.2.1.1 Formulation and processing. The ceramic component shall be homogenous and of one piece (monolithic) construction. All ceramic components shall be manufactured in distinct production lots (see 4.4.1) and a sequential marking system of individual components shall be provided (see 3.6.1). The contractor shall not change the ceramic formulation or processing technique without approval of the contracting officer. Information shall also be

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furnished as to what production lots are affected by the change in ceramic formulation or processing technique. A change in the ceramic formulation or processing technique shall constitute a new lot.

3.2.2 Fiberglass laminate. The fiberglass laminate shall consist of a minimum of 20 percent by weight of unsaturated polyester type, unfilled resin and may contain modifying monomers or polymers and the remainder shall be undyed fiberglass fabric. Resin content shall be determined as specified in 4.4.2.2. The fiberglass laminate shall be self-extinguishing or non-burning when tested for flammability as specified in 4.4.2.2.

3.2.3 Cloth, nylon. The nylon cloth for the spall cover shall conform to class 1 of MIL-C-12369. The color shall be natural.

3.2.4 Rubber edging. The rubber edging shall be made of black material conforming to grade, type, class, and suffix requirements of 3AA615A14F17Z of ASTM D 2000. The rubber edging, when used in conjunction with the adhesive specified in 3.2.6.2 and procedure specified in 3.4.4 shall be capable of meeting the adhesion requirement specified in 3.5.3 (see 6.4.1, 6.4.2, and 6.5).

3.2.5 Vest. The vest shall conform to MIL-V-43544 in the applicable size.

3.2.6 Adhesives.

3.2.6.1 For bonding ceramic to fiberglass laminate. The adhesive for bonding the ceramic component to the fiberglass laminate component shall be solventless liquid or film epoxy type, two-part polysulfide type, or two-part polyurethane type and shall be capable of passing the requirement in 3.5.2 (see 4.4.2.1).

3.2.6.2 For bonding spall cover and rubber edging to inserts. The adhesive for bonding the spall cover and the rubber edging to the inserts shall be a natural or synthetic rubber base curing type (see 6.5) capable of passing the adhesion requirements specified in 3.5.3 and 3.5.4 (see 4.4.2.1).

3.2.6.3 Alternate materials and adhesive systems. Alternate materials and adhesive systems may be used in the fabrication of the armor when approved by the contracting officer and when demonstrated to be equal or better in performance than what is specified above. Complete material analysis and test data which will substantiate the above shall be furnished. For alternate adhesives to bond the ceramic and fiberglass components, test data shall include results of ballistic and temperature extremes testing.

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3.3 Design. The body armor shall consist of a fiberglass laminate component bonded to the back of a ceramic component forming a composite armor insert. The exposed surface of the ceramic component shall be covered with a spall cover. A rubber edging shall extend around the entire periphery of the composite armor insert. The insert shall conform to the specified class and size. The front insert shall be fabricated in accordance with Drawing 8-2-217, and the back insert in accordance with Drawing 8-2-218. The finished dimensions on the drawings apply before application of spall cover and rubber edging. Approved manufacturer's drawings for ceramic components may be used in lieu of the drawings specified above (see 6.2).

3.4 Construction.

3.4.1 Ceramic component. The ceramic component shall conform to the requirements of 3.2.1 and 3.2.1.1 and shall be uniform in thickness within 5 percent when examined as specified in 4.4.3.1. Patching of the ceramic component is permitted in the green state only before oven or kiln firing. The ceramic component shall be free from manufacturing anomolies or discontinuities when inspected as specified in 4.4.3.2, 4.4.3.3 and 4.4.5.2.

3.4.2 Fiberglass laminate component. The fiberglass laminate component shall be fabricated from the materials specified in 3.2.2. The finished component shall be free from blisters, torn fabric, dents, crazing and surface roughness. Glass fibers protruding along the edges shall be removed.

3.4.2.1 Molding. The resin treated fabric shall be molded by vacuum or pressure bag technique (see 6.6) or matched metal molds into the required configuration without breaking or damaging the glass fiber. Such procedures and methods shall be capable of yielding uniform properties in the completed structure. The fiberglass laminate may be molded and attached to the ceramic component in one operation if performance requirements are maintained. No patching shall be performed after the material has been molded.

3.4.3 Spall cover. The nylon spall cover shall be fabricated from one ply of the cloth specified in 3.2.3 and shall cover the entire outer surface of the ceramic component except that the cover shall finish even or within 1/8 inch of the edge around the entire periphery.

3.4.4 Assembly of components by bonding. All components to be assembled shall be thoroughly cleaned of all foreign matter. Surfaces to be bonded shall be properly prepared in a manner which will insure a proper bond capable of meeting the applicable performance requirements specified in 3.5. The required adhesive shall be applied uniformly over the entire contact areas of the components to be joined. There shall be no evidence of unadhered areas and no smearing of the adhesives on the exposed surfaces of the components, except when bonding the rubber edging, the adhesive may extend beyond the edges of the rubber edging up to a maximum of 3/16 inch.

3.5 Performance.

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3.5.1 Ballistic resistance. The body armor insert without spall cover and rubber edging shall have a V_{50} protection ballistic limit of not less than 2850 feet per second with a caliber .30 AP M2 projectile when tested as specified in 4.4.3.5.

3.5.2 Temperature resistance. There shall be no delamination between the fiberglass laminate component and the ceramic component or between the fiberglass plies when tested as specified in 4.4.3.5.

3.5.3 Adhesion of rubber edging. There shall be no visual separation of the rubber edging from the armor insert when tested as specified in 4.4.6.

3.5.4 Adhesion of the spall cover. There shall be no visual peeling of the tab formed on the spall cover when tested as specified in 4.4.6.

3.5.5 Areal density. The areal density of the body armor inserts, without spall cover and rubber edging, when tested as specified in 4.4.3.5, shall be as follows for the applicable class:

Class 1 - 9.0 $\pm$ 0.2 pounds per square foot

Class 2 - 8.0 $\pm$ 0.2 pounds per square foot

Class 3 - 7.5 $\pm$ 0.2 pounds per square foot

Class 4 - 6.9 $\pm$ 0.2 pounds per square foot

3.6 Identification marking.

3.6.1 Ceramic component marking. The serial number shall be marked on the outside face surface within 1 inch of the lower edge of each ceramic component, before it is fired by inscribing, embossing, or with high temperature resistant inorganic ink or after firing with a permanent marking to a maximum depth of 0.010 inch so that the serial number will be legible when the spall cover is stripped by mechanical or solvent action.

3.6.2 Completed armor insert marking. The manufacturer's code number, contract number, lot number, serial number, and date of manufacture, as well as the nomenclature, federal stock number, class, and size of the armor and designation of the armor as to being a front or back insert shall be permanently and legibly marked on the center of the face surface of each spall cover, at the lower end of the insert approximately 1 inch above the rubber edging, in characters 1/4 inch high minimum. The serial number shall be the same as that on the ceramic component. The letters "U.S." shall be applied approximately 2 inches above the identification markings, centered on the spall cover, in characters 1 inch high minimum. On the upper half of the spall cover, the words "DO NOT DROP" shall be printed in characters 1 inch high minimum. The above marking shall conform to type IV, class 9 of DDD-L-20. The fastness for the

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class 9 label shall be as specified for class 5 labels. A list in duplicate of the serial numbers used for each armor production lot shall be furnished the contracting officer at time of delivery.

3.7 Production data. The following information determined during production of the armor shall be made a matter of record and shall be furnished to the contracting officer upon request. This data shall be identified with the serial number of the armor insert.

a. The weight of each ceramic component.
b. The weight of each composite body armor insert without spall cover and rubber edging.
c. All thickness measurements taken for each ceramic component.

3.8 Workmanship. The body armor shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.1.3 Radiographic and ballistic laboratory approval. Radiographic examination and ballistic testing shall be performed in certified laboratories (see 6.9). The radiographic equipment and procedures shall be in accordance with Appendix B. Ballistic testing equipment and procedures shall be in accordance with Appendix A.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Inspection lots. The maximum lot size for all tests and examinations shall not exceed 1000 items of either front or back ceramic component or composite armor inserts, as applicable, sequentially produced. Lots shall be composed of front ceramic components or composite armor inserts only or back ceramic components or composite armor inserts only.

4.4.1.1 Initial production lot. The minimum lot size for the initial production lot shall be as specified in the contract or order (see 6.2).

4.4.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2.1 Certification. The contractor shall furnish a certificate of compliance certifying the following:

a. That the ceramic formulation and/or production processing have not been changed from that previously submitted to the contracting officer and that the ceramic inserts are homogeneous and manufactured in distinct production lots as specified in 3.2.1.1.

b. Adhesive for bonding ceramic component to fiberglass laminate component conforms to 3.2.6.1 or 3.2.6.3.

c. Adhesive for bonding spall cover to ceramic component conforms to 3.2.6.2 or 3.2.6.3.

d. Adhesive for bonding rubber edging to composite laminate conforms to 3.2.6.2 or 3.2.6.3.

4.4.2.2 Testing of fiberglass laminate component. Testing shall be performed on the fiberglass laminate component for the characteristics listed in table I. The lot size shall be expressed in units of front or back fiberglass laminate components. The sample unit shall be one fiberglass laminate component. The

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inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5. When the fiberglass laminate is molded and bonded to the ceramic component in a simultaneous operation, the specimens required for testing shall be cut from composite armor inserts that have been ballistically tested.

TABLE I. Testing of fiberglass laminate component

Characteristic	Requirement paragraph	Test method	No. of det. per sample unit	Results reported as
Resin content (percent)	3.2.2 1/	ASTM D 2584	3	Avg. of 3 det. to nearest 1 percent
Flame resistance	3.2.2 1/	ASTM D 635	1	Pass or fail

1/ A certificate of compliance will be acceptable for the stated requirement after the first acceptable production lot.

4.4.3 In-process inspection. Examination shall be made during the manufacturing process for the requirements listed below. Whenever nonconformance is noted, correction shall be made to the items affected, the lot in progress, and the operation. Items which cannot be corrected shall be removed from production.

Check for	Component or requirement
Proper preparation, cleanness, and uniformity of application of adhesive.	Surfaces of ceramic and fiberglass laminate components prior to bonding. Surfaces of rubber edging to be bonded. Surfaces of ceramic and fiberglass laminate components where rubber edging is to be bonded.
Size of cut parts	Spall cover (after attachment to armor).

4.4.3.1 Examination of ceramic component for thickness. The ceramic component shall be examined for thickness prior to bonding to the fiberglass laminate. One hundred percent inspection shall be performed, and any ceramic component which fails the examination shall be rejected. The ceramic component shall be uniform in thickness as specified in 3.4.1 when examined in the following manner. The thickness shall be measured to 0.001 inch at five locations on the ceramic component. With the ceramic component considered as divided into four quadrants (one for each corner area), one reading shall be taken within each quadrant. Each reading shall be taken a minimum of 1-1/2 inches away from the edge. In addition, one reading shall be taken at the

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center of the ceramic component. All readings shall be spaced a minimum of 3 inches apart. The five readings shall be averaged and reported to the nearest 0.001 inch as the overall average. The difference between the greatest value and the overall average and the smallest value and the overall average shall not be more than 5 percent of the overall average thickness. All determinations shall be reported with the overall average. A deep throat type depth gage conforming to Federal Products Corp. Model KP 125 or KP 129 or equal with standard measuring tips, shall be used for taking the measurements.

4.4.3.2 Liquid penetrant and visual examination of ceramic component. The ceramic component shall be examined visually and with liquid penetrant on both sides and edges, after furnace firing and prior to application of the adhesive for bonding of the ceramic component to the fiberglass laminate component. Unless otherwise specified the examination shall be performed at the contractor's plant (see 6.2). The liquid penetrant examination shall be performed in accordance with Appendix C, visual examination shall be performed in accordance with Appendix D. One hundred percent examination shall be performed on all ceramic components in the lot (see 4.4.1). Defects shall be classified in accordance with the applicable defects listed in table II. Any ceramic component found with one or more major defects or four or more minor defects shall be rejected.

4.4.3.3 Visual examination of composite armor insert. Visual examination shall be conducted in accordance with Appendix D, on the face side and edge of the ceramic portion of the composite armor insert, and on the exposed surface and edge of the fiberglass laminate component just prior to application of the adhesive for bonding of the spall cover. One hundred percent examination shall be performed on all composite armor inserts in the lot (see 4.4.1). Defects shall be classified in accordance with the applicable defects listed in table II. Any composite armor insert found with one or more major defects or four or more minor defects shall be rejected.

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TABLE II. Classification of defects applicable to ceramic component by liquid penetrant and visual examination; defects applicable to composite armor insert by visual examination; defects applicable to completed armor insert by radiograph examination

Examine	Defect	Defects applicable to examination by: V-Visual	L-Liquid penetrant	R-Radiographic 1/	Classification Major	Classification Minor
Ceramic component	a. Edge crack (filled or unfilled) greater than 1-1/2 inches in length and through thickness of plate 2/, 4/, 10/	V	L	R	X	
	b. Edge crack (unfilled) greater than 3/4 inch in length and not through thickness 2/, 4/, 5/, 10/	V	L	R		X
	c. Edge crack (unfilled) less than 3/4 inch in length and through thickness 2/, 4/, 5/, 10/	V	L	R		X
	d. Any crack (unfilled) over 1/2 inch in length other than edge crack 3/, 4/, 10/	V	L	R	X	
	e. Any crack (filled) other than edge crack and over 1/2 inch in length 7/, 8/	V	L	R	X	
	f. Laminar crack along edge greater than 3/4 inch in length 10/	V	L	--	X	

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TABLE II. Classification of defects applicable to ceramic component by liquid penetrant and visual examination; defects applicable to composite armor insert by visual examination; defects applicable to completed armor insert by radiograph examination (cont'd)

Examine	Defect	Defects applicable to examination by: V-Visual	L-Liquid penetrant	R-Radiographic 1/	Classification Major	Classification Minor
Ceramic component (cont'd)	g. Shrinkage, filamentary type, (filled or unfilled) indications: 4/, 7/, 8/					
	(1) Three or less such defects in any 2 inch diameter circle anywhere on the plate whose combined length, when measured from end to end in a straight line, is greater than 1 linear inch.	--	--	R	X	
	(2) Four or more such defects regardless of length in any 2 inch diameter circle anywhere on the plate.	--	--	R	X	
	h. Sponge area, (unfilled) -over 0.2 inch^2 (1/2 diameter circle) per any 2 inch diameter circle 4/, 7/, 8/	V	L	R	X	
	i. Silicon rich area -over 0.2 inch^2 (1/2 diameter circle) per any 2 inch diameter circle 4/, 7/, 8/	--	--	R	X	

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TABLE II. Classification of defects applicable to ceramic component by liquid penetrant and visual examination; defects applicable to composite armor insert by visual examination; defects applicable to completed armor insert by radiograph examination (cont'd)

Examine	Defect	Defects applicable to examination by: V-Visual	L-Liquid penetrant	R-Radiographic 1/	Classification Major	Classification Minor
Ceramic component (cont'd)	j. Any internal void or unfilled area greater than 0.2 inch^2 per any 2 inch diameter circle 4/, 10/	--	--	R	X	
	k. Pit greater than:					
	(1) 0.030 inch in depth and diameter but not more than 0.060 inch in depth 5/, 6/, 10/	V	L	R		X
	(2) 0.030 inch in diameter and greater than 0.060 inch in depth 10/	V	L	R	X	
	(3) 3/16 inch in the longest lateral direction but not more than 5/16 inch and 0.030 inch or less in depth 5/, 6/, 10/	V	L	R		X
	(4) 5/16 inch in the longest lateral direction and 0.030 inch or less in depth 10/	V	L	R	X	
	l. Rivelet over 1-1/2 inch in length 9/	V	--	R	X	

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TABLE II. Classification of defects applicable to ceramic component by liquid penetrant and visual examination; defects applicable to composite armor insert by visual examination; defects applicable to completed armor insert by radiograph examination (cont'd)

Examine	Defect	Defects applicable to examination by: V-Visual	L-Liquid penetrant	R-Radiographic 1/	Classification Major	Classification Minor
Ceramic component (cont'd)	m. Edge chip on either surface beyond 1/8 inch outer boundary:					
	(1) Greater than 3/8 inch in any lateral direction but not more than 3/4 inch and 1/16 inch or less in depth 5/, 10/	V	L	R		X
	(2) Greater than 3/4 inch in any lateral direction and 1/16 inch or less in depth 10/	V	L	--	X	
	(3) Greater than 1/16 inch in depth but not more than 1/8 inch 5/, 10/	V	L	--		X
	(4) Greater than 1/8 inch in depth 10/	V	L	--	X	
	n. Edge chip on either surface within the 1/8 inch outer boundary greater than 3/16 inch in depth 10/	V	L	--	X	

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TABLE II. Classification of defects applicable to ceramic component by liquid penetrant and visual examination; defects applicable to composite armor insert by visual examination; defects applicable to completed armor insert by radiograph examination (cont'd)

Examine	Defect	Defects applicable to examination by: V-Visual	L-Liquid penetrant	R-Radiographic 1/	Classification Major	Classification Minor
Ceramic component (cont'd)	o. Scratch or gouge:					
	(1) 0.030 to 0.060 inch in depth 5/, 6/, 10/	V	L	--		X
	(2) Greater than 0.060 inch in depth 10/	V	L	--	X	
	p. Any mend or patch 10/	V	--	--	X	
	q. Markings (serial number) omitted, not located or applied as specified 10/	V	--	--		X
Fiberglass laminate component	a. Any delamination from the ceramic component 10/	V	--	--	X	
	b. Exposed fibers on edge 10/	V	--	--		X
	c. Any blister, crack, torn fabric, crazing, dents or surface roughness 10/	V	--	--		X
	d. Any delamination of fiberglass plies 10/	V	--	--		X

1/ Radiographic examination shall be performed using the composite armor insert.

2/ Edge cracks are defined as those within the 1-1/2 inches boundary around the periphery of the ceramic component.

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TABLE II. Classification of defects applicable to ceramic component by liquid penetrant and visual examination; defects applicable to composite armor insert by visual examination; defects applicable to completed armor insert by radiograph examination (cont'd)

3/ Cracks other than edge cracks are defined as those beyond the 1-1/2 inch edge boundary or those that start within the 1-1/2 inch edge boundary and extend beyond the 1-1/2 inch boundary.

4/ For the purposes of radiographic examination and interpretation, the following definitions or descriptions shall apply (see 6.8):

Cracks. Well-defined line or lines having sharp terminal points indicating a break in the ceramic material.

Shrinkage filamentary type indications. Sharply defined wavy or irregular line or lines usually less than 3/4 inch in length occurring within the ceramic material and indicating an internal discontinuity.

Internal voids or unfilled areas. Round or elongated, smooth edged, dark spots occurring individually or randomly distributed within the ceramic material.

Sponge. Dark localized area or areas occurring within the ceramic and indicating low ceramic density or high porosity.

Silicon Rich Area. A lighter localized area or areas indicating a high concentration of silicon within the ceramic.

5/ All such surface minor defects so identified in the visual and liquid penetrant examinations shall be described, located, and annotated so that they will not be identified as internal voids or unfilled areas (defect h) when and if the ceramic component is subjected to radiographic examination.

6/ Any two minor defects on the surfaces of the ceramic component, except within the 1-1/2 inch edge boundary, less than 3/8 inch apart shall be classified as a major defect.

7/ Applies to silicon carbide type only.

8/ Applies to boron carbide/silicon carbide/silicon type only.

9/ Applies to boron carbide type only.

10/ Applies to all classes of ceramic.

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4.4.3.4 Dimensional examination of composite armor inserts prior to application of spall cover. The composite armor insert shall be examined prior to application of spall cover for conformance to the dimensional requirements specified on the applicable drawing. Any dimension that is not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of composite armor inserts, front or back without spall cover (see 4.4.1). The sample unit shall be one composite armor insert, front or back. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.3.5 Testing of composite armor inserts prior to application of spall cover. The composite armor inserts shall be tested for the characteristics listed in table III. The lot size shall be expressed in units of front or back composite armor inserts prior to application of spall cover (see 4.4.1). The sample unit for ballistic and areal density tests shall consist of 10 randomly selected composite armor inserts and for the temperature resistance test the sample unit shall be one composite armor insert. When more than one size is in the lot, the sample unit used in the ballistic test shall contain at least two of each size. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5. All areal density and ballistic tests shall be witnessed by Government inspector.

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TABLE III. Composite armor insert tests

Characteristic	Requirement paragraph	Test method	Requirements applicable to	No. deter. per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Ballistic resistance 2/	3.5.1	4.5.1	Sample unit	1 1/	--	Ft./sec.
Temperature resistance	3.5.2	4.5.2	Sample unit	1	X	--
Areal density 2/	3.5.5	4.5.5	4/	3/	--	0.1 lb./sq.ft.

1/ The insert having the lowest area density shall be tested first. Each succeeding insert tested shall be the next insert having the lowest areal density.

2/ Areal density test and ballistic test shall be conducted on the same samples.

3/ Areal density to be performed on each armor insert of the sample unit.

4/ Requirements applicable to individual inserts. All thickness determinations shall be reported.

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4.4.4 End item visual examination. The body armor inserts shall be examined for the defects listed below. The lot size shall be expressed in units of front or back armor inserts (see 4.4.1). The sample unit shall be one armor insert, front or back. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Spall cover	Cut, hole, tear, or abraded area Any wrinkle or crease Any mend or patch Any areas not bonded to ceramic component exceeding 1/4 inch in longest lateral direction
Rubber edging	Not continuous Any dimension on height of edging, size and depth of cut-outs at radii points, or space between butted ends exceeds requirements Torn or cut except where permitted Does not overlap peripheral edge of armor as specified Any area of non-adherence in excess of 1/2 inch
Cleanness	Grease or oil stains clearly noticeable Adhesive smeared on spall cover or rubber edging (except where allowable)
Construction and workmanship	Any component missing or not assembled as specified (unless otherwise classified herein) 1/
Identification markings	Not as specified, missing, of improper size, or illegible

1/ Presence of this defect in one or more of the samples shall be cause for rejection of the lot.

4.4.5 End item radiographic examination. The completed armor inserts shall be radiographically examined. Industrial radiography shall be performed in accordance with Appendix B. Radiography shall include the entire ceramic component area and shall contain the same permanent marking as the ceramic component. Radiographic films shall be made available to the Government upon request. The radiographic equipment and procedures shall be in accordance with MIL-STD-453, except where noted in Appendix B.

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4.4.5.1 Initial production lot. One hundred percent radiographic inspection of the initial lot of completed armor inserts shall be conducted (see 4.4.1.1). Defects shall be classified in accordance with the applicable defects in table II. For each completed armor insert the acceptance number for major defects shall be zero and for minor defects three. Any completed armor insert found with one or more major defects or four or more minor defects shall be rejected.

4.4.5.2 Production lots after initial lot. All production lots after the initial lot shall also be radiographically inspected. The lot shall be expressed in terms of front or back armor inserts (see 4.4.1). The sample unit shall be one armor insert, front or back. The inspection level shall be II. The acceptable quality level shall be 0.25 percent defective items. A defective item is an armor insert containing one or more major defects or four or more minor defects when examined for the applicable defects listed in table II.

4.4.6 End item testing. The body armor inserts shall be tested for the characteristics listed in table IV. The lot size shall be expressed in units of front or back armor inserts (see 4.4.1). The sample unit shall be one armor insert. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE IV. End item tests

Characteristic	Requirement	Test method	No. deter. per sample unit	Results reported as pass or fail
Adhesion of rubber edging	3.5.3	4.5.3	2	X
Adhesion of spall cover	3.5.4	4.5.4	1	X

4.4.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flap, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container.

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Examine	Defect
Content	Number per container is more or less than specified.

4.5 Methods of inspection.

4.5.1 Ballistic test for armor. The ballistic test shall be conducted in accordance with appendix A unless otherwise specified. The areal density shall be reported for each test insert.

4.5.2 Temperature extremes test. The armor insert, less the rubber edging, and spall cover, shall be heated in an oven operating at 160° ± 5°F for 6 hours, ± 1/4 hour. The test specimen shall be then allowed to cool to room temperature followed by cold exposure at -65° ± 5°F for 6 hours, ± 1/4 hour. The specimen shall be allowed to warm to room temperature (70° to 90°F) and shall then be examined at edges for evidence of delamination of the fiberglass laminate component from the ceramic component or any delamination of the fiberglass plies. Specimens that have passed the test may be returned to the production lot.

4.5.3 Adhesion test of rubber edging. The rubber edging shall be tested for adhesion by application of a 5 pound minimum static shear load. The 5 pound load shall be applied to the rubber edging using a clamp device conforming to that shown on figure 1. With the armor insert vertically suspended or secured, the clamp shall be adjusted such that the contact points evenly make contact with the top edges of the rubber edging, see figure 1. The clamp shall not come in contact with the spall cover or fiberglass laminate surface of the armor insert. The load shall be applied to all four sides of the back armor insert and to the three sides (left, right, and bottom) of the front insert in the approximate center of each side. Each location shall be tested separately and the load shall be applied for a minimum of 10 minutes. Any evidence of separation of the edging from the armor insert shall constitute failure of the test.

4.5.4 Adhesion test for spall cover. The sample for this test may be made from an armor insert which has been tested ballistically using an undamaged area, or a small test specimen made identically to the armor insert with respect to surface preparation, adhesive application, and bonding. Make parallel cuts through the spall covering 4 inches long and 1 inch apart along the weave. At one end of the cut, make a cut at a right angle so that a 1-inch wide by 2-inch long tab can be peeled from the armor insert. Attach a 2-pound weight to this tab so that the resultant peel force is normal to the plane of the armor insert. Make gage marks on the specimen for the measurement of the amount of peeling during the test. Allow the weight to hang for 4 hours minimum at room temperature (70° to 90°F). Any visual peeling of the tab exceeding 1/16 inch constitutes failure of the test.

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4.5.5 Areal density. The areal density in pounds per square foot of the composite armor insert, without spall cover and without rubber edging, shall be calculated from measurements on the composite armor insert. The composite armor insert shall be weighed to the nearest 0.01 pound. The thickness shall be measured to 0.001 inch at four corner locations on the composite armor insert. Six measurements shall be made at each corner with the first measurement taken approximately 1 inch from the edge of the corner and the remaining five taken on a straight line toward the center of the armor with the measurements spaced at approximately 1-inch increments. Three measurements shall be taken at the low points of the fabric weave and three at the high points. A deep throat type depth gage conforming to Federal Products Corp. Model KP125 or KP129, or equal, with standard tips, shall be used for taking measurements. The average of the 24 readings shall be used in the calculation of the areal density. The edges of the glass fiber laminate shall be painted with a slow drying lacquer to seal the edges. The lacquer shall be allowed to dry (at least 1/2 hour) before proceeding with the determination of density. The average density of the composite armor insert shall be determined by Method A-3 of ASTM D 792, except the immersed weight shall be determined to 0.01 pound and the liquid used shall be clean tap water, containing a wetting agent if necessary, which shall be changed just prior to examination of each lot. The areal density of the composite armor insert without spall cover and rubber edging shall be calculated to the nearest 0.01/lb/sq. ft. as follows:

Areal density (lb/sq. ft.) = 5.202 X Density (g/cc) X Thickness (in.)

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. The insert(s) (type I and II) shall be inserted in the appropriate pocket of the vest and the flaps securely fastened. Each body armor assembly (type I, II, III and IV) shall be completely wrapped in not less than 1-inch cushioning material conforming to type II, class B of PPP-C-843 or padding conforming to type I, grade A, class 1 or 2 of CCC-P-91. The cushioning or padding material shall be secured with pressure-sensitive tape or by means of cross-tying with cotton tape or twine. The front and back armor assembly (type II) shall be positioned as worn (front over back) prior to securing the cushioning material in order that the two sections can be taped or tied together. Two type III or two type IV body armor of one class and size only, shall be centrally positioned on opposite sides of a full size fiberboard pad. The inserts shall have the respective concave sides facing the pad and be alternately reversed top to bottom. The pad shall conform to type CF (variety TW), class domestic of PPP-F-320. The inserts shall be secured to the pad by cross-banding with filament reinforced pressure-sensitive tape or by cross-tying with cotton tape or twine.

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5.1.1.2 Intermediate packing. Two of type I, III, or IV or one of type II body armor of one class and size only, unit packed as specified in 5.1.1.1, shall be further packed flat in a snug-fitting intermediate fiberboard box conforming to style RSC-L, grade V3c of PPP-B-636. The inside of each box shall be fitted with a box liner and top and bottom pads conforming to type CF (variety TW), class domestic of PPP-F-320. The height of the box liner shall be the inside depth of the box less the combined thicknesses of the top and bottom pads. The top and bottom pads shall measure 1/4 inch less than the inside length and width dimensions of the box. All void spaces around the body armor shall be filled with the same cellulosic cushioning or padding specified in 5.1.1.1 in a manner to immobilize the contents from movement while in transit. Each fiberboard box shall be closed and water-proofed with tape in accordance with the appendix of PPP-B-636.

5.1.2 Commercial. Body armor shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Eighteen type I, III, or IV, or nine type II body armors of one class and size only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. Level A intermediate packs shall be packed flat, three in length, one in width, and three in depth within a shipping container. Each shipping container shall be provided with skids as specified in the container specification. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601. Toward the end of the contract or when there are less than the required amount per container of the same type, class and size, mixed sizes of one type and class only may be packed within the same container.

5.2.2 Level B packing. Eighteen type I, III, or IV, or nine type II body armors of one class and size only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to domestic type, style A or I, grade B, type 2 load of PPP-B-601. Level A intermediate packs shall be packed flat, three in length, one in width, and three in depth within a shipping container. Each shipping container shall be provided with skids as specified in the container specification. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601. Toward the end of the contract or when there are less than the required amount per container of the same type, class, and size, mixed sizes of one type and class only may be packed within the same container.

5.2.3 Commercial packing. Body armor, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, intermediate packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

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5.3.1 Special marking. Each intermediate pack and shipping container shall be provided with fragile labels in accordance with MIL-STD-129 and in addition, the top panel of each package and container shall be printed or stenciled in black ink with the following legend:

"DO NOT DROP"

5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes of one type and class only shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED NSNs" plainly stamped or printed thereon and under these words shall be legibly printed the correct quantity of NSNs contained therein.

6. NOTES

6.1 Intended use. The body armor is intended for use by aviation personnel to provide protection against small arms fire.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this document.
b. Type, class, and size required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. List of approved manufacturer's drawings for ceramic components of body armor inserts (see 3.3).
e. Minimum size of initial production lot (see 4.4.1.1)
f. Location for performance of liquid penetrant on the ceramic component (see 4.4.3.2).
g. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting offer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Rubber edging.

6.4.1 Application. Suppliers are advised to be careful in the selection of the edging to be used and in the conduct of bonding operations. One prime factor in obtaining good adhesion is that the content of plasticizers, softeners, and extenders in the edging material be kept low. It is important also that the specified abrading be done across the full width of the edging; for this purpose, it is suggested that the edging channel be opened up flat and that a fine wire wheel (0.004-inch diameter wire) of 1-1/2 inch minimum width be used across the full width of the edging channel. Solvent cleaning is preferably done by a cloth wipe than by flooding.

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6.4.2 Source. Rubber edging has been obtained from Pawling Rubber Company, Pawling, NY, which in conjunction with adhesive EC 1300 (see 6.5), has met all adhesion requirements of this specification.

6.5 Adhesive for rubber edging and spall cover. Minnesota Mining and Manufacturing Company adhesive EC 1300 has been used in the manufacture of prototype items to adhere the rubber edging and spall cover.

6.6 Fabrication of fiberglass laminate. Fabrication of the fiberglass laminate has been made by the following technique. One piece pressed ceramic components were used as molds. The fabrication technique consisted of hand laying the polyester impregnated glass cloth into the ceramic cavity. The mold was fitted with a polyvinyl alcohol (PVA) vacuum bag and the assembly placed in an autoclave. The evacuated assembly was pressurized at 100 pounds per square inch at room temperature for 2 hours and then subjected to a temperature of 200°F for 3 hours without removing the pressure. The assembly was removed from the autoclave and allowed to cool to room temperature. The ceramic was removed and the front of the fiberglass laminate backing abraded to provide good adhesion when cemented to the ceramic piece.

6.7 Mold release agents. Manufacturers are advised not to use mold release agents that are difficult to wash off or which are known to interfere with adhesive bonding even in small residual amounts.

6.8 Definition photographs. Photographs showing examples of the definitions defined in footnote 4 of table II may be obtained from the procuring office issuing the invitation for bids.

6.9 Certification activity. Unless otherwise designated, the Army Materials Technology Laboratory, Watertown, MA, is responsible for the certification of radiographic examination facilities and procedures and for the certification of ballistic testing facility and equipment.

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.11 Supersession data. This document includes the requirements of LP/P DES 1-82, dated 18 January 1982.

6.12 Subject term (key word) listing.

Body Armor
Armor, body
Protection, small arms
Ceramic
Aircrew

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6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 11

Review activities:

Army - MR
Air Force - 99
DLA - CT

User activities:

Navy - MC, SH

Preparing activity:

Army - GL

Project No. 8470-0112

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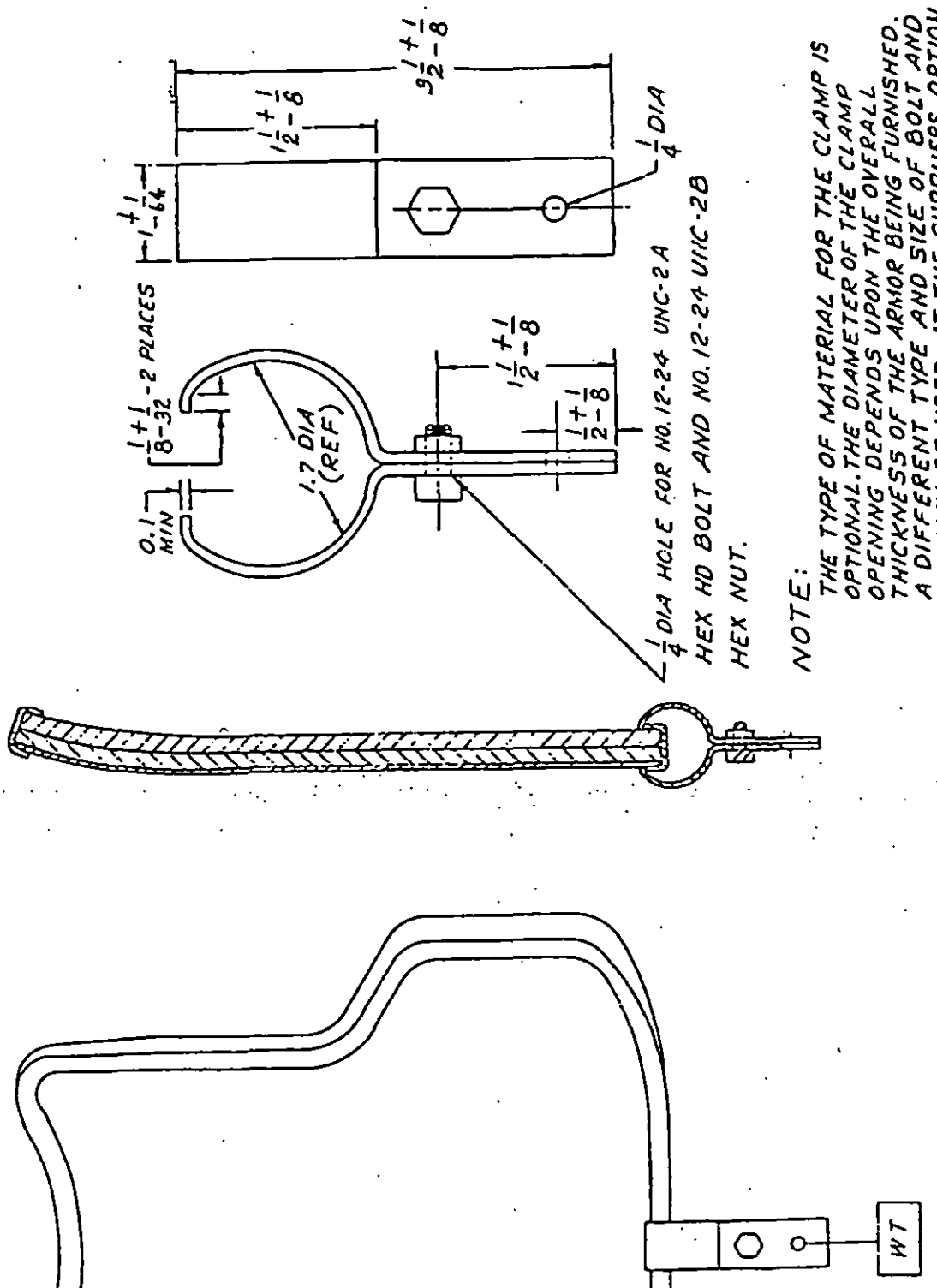


FIGURE 1. CLAMP FOR APPLICATION OF WEIGHT IN PERFORMING RUBBER EDGING ADHESION TEST, AIRCREWMAN'S BODY ARMOR.

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APPENDIX A

Ballistic Test

10. SCOPE

10.1 Scope. This appendix details the procedure to be used in determining the ballistics resistance of ceramic faced armor materials with caliber .30 AP M2 projectiles. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

DRAWINGS

Frankford Arsenal

C6138195 - Bullet

30. REQUIREMENTS

30.1 Equipment. Equipment for ballistic testing shall consist of the following items:-

a. Gun Mount - The gun mount shall be suitable for firing the caliber.30 gun to muzzle velocity of at least 3000 ft/sec.

b. Test Sample Mounting - The armor test sample shall be mounted in a frame which is suspended from overhead by means of steel cables, wires, or chains. The armor test sample shall be secured in the vertical position, perpendicular to line-of-flight projectile, using quick release clamps. The frame supports and clamps must be capable of retaining the sample and withstanding shock resulting from ballistic impact by caliber .30 AP M2 projectile (kinetic energy approximately 3500 ft. -lbs.) on the test sample. The test sample mounting shall be capable of adjustment for moving the sample in the vertical or horizontal directions so that the point of impact can be located anywhere on the sample, and rotation on the vertical axis so that zero degree obliquity impacts can be achieved anywhere on the sample.

c. Weapon - A caliber .30 weapon shall be used.

d. Projectile - The projectile to be used is the caliber .30 AP M2 standard service projectile (bullet) conforming to Frankford Arsenal Drawing C6138195.

e. Witness Plate - The witness plate shall be a 2024-T3 or 2024-T4 aluminum alloy sheet, 0.020 inch thick and shall be a minimum of 11 inches by 14 inches in size.

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APPENDIX A

f. Velocity measuring equipment.

1. Chronograph - an electronic counter type chronograph measuring to the nearest microsecond or as a minimum to the nearest 10 microseconds.

2. Detectors - Either high-velocity luminline screens, or electrical contact screens which either open or close an electric circuit by passage of the projectile through the detector. Contact screens may consist of metallic foils separated by a thin insulating layer, or may consist of a circuit printed on paper with the circuit spacing such that the projectile passing through the screen will "break" the circuit.

g. Propellant - Any propellant which is standard for the weapon may be used, however, when other than a standard propellant is used, extreme care must be exercised in selecting the charge weight. A projectile velocity - propellant charge curve for the weapon shall be determined before any testing is performed. This curve is required to provide a basis for selecting a powder charge to achieve a desired velocity. It is recommended that the propellant storage and weighing area be maintained at $72^{o} \pm 5^{o}F$ and 50 ± 5 percent relative humidity.

30.2 Set-up. The required set-up for the ballistic equipment is shown in figure 2.

30.3 Data sheet. Data sheets shall be provided and shall contain the following information:

a. Item document number.
b. Material description (including physical characteristics if required).
c. Material identification number for each test sample.
d. The areal density of each test sample.
e. Temperature and humidity of the test facility.
f. Date.
g. Personnel conducting test and any witnesses.
h. Weapon used.
i. Projectile used.
j. Projectile weight, grains.
k. Type of propellant.
l. Weight of propellant for each shot, grains.
m. Impact velocities used in computing V_{50}'s with highest partial penetration, lowest complete penetration, range (spread), and velocities of rounds all being listed.
n. Witness plate characteristic, partial or complete.
o. Calculated ballistic limit (V_{50}).
p. Any remarks pertinent to the conduct of the test, or behavior of the material.

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APPENDIX A

30.4 Procedure.

30.4.1 Marking, samples. Areal density must have been previously determined on the test samples before they are tested ballistically. Each test sample will be marked in the following manner: (See figure 3)

a. The approximate center of the test sample will be located.

b. From the center of the test sample, a line is drawn to the center of the bottom of the test sample.

c. From the center of the test sample, lines are drawn on both sides of the test sample to the nearest edge (the narrowest) portion of the test sample. (see Figure 3)

d. The upper area is marked A, the lower right area B, and the lower left area C, respectively.

e. The first test sample will be impacted in area A, the second test sample in area B, and the third test sample in area C. Each successive test sample will be fired in the same order, i.e. area A, B, and C. Only one shot will be fired in each test sample. All shots shall be fired at the approximate center of the area to be impacted.

30.4.2 Velocity, projectile. A test round will be fired through a witness plate to determine the exact position of impact. Three additional rounds shall be fired using the appropriate projectile velocity - propellant charge curve for the weapon being used, and selecting propellant weights to achieve a velocity of approximately 2850 ft/sec. The last three of these rounds shall be within the range of 2850 to 2900 ft/sec (hereinafter designated the "reference velocity"). If it is not, additional rounds shall be fired and the projectile velocity - propellant charge curve corrected until reference velocity is achieved. The propellant charge for the last warm-up round shall be considered the reference charge for the test. From the curve, the increments to and decrements from the reference propellent charge to yield approximate velocity changes at the reference velocity of 100 ft/sec and of 50 ft/sec shall be determined and recorded. These shall be the only increments used during the test. The reference charge and propellant increments and decrements may be redetermined between tests whenever difficulty is experienced in completing the last test. The entire procedure must then be repeated. All reference velocities, and propellant charges, and increments and decrements of propellant weights developed for a test or utilized from a previous test shall be recorded. All the records for the procedure shall be recorded with the test that immediately follows. This procedure must be repeated for succeeding tests if the weapon has not been fired within two hours previous to the test.

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30.4.3 Mounting, sample. The sample will be mounted as specified in 30.1 b, Test Sample Mounting. The point of impact will be located on the test sample as specified in 30.4.1 and will be positioned to lined up with the previously determined line of flight. The point of impact is now checked for obliquity. The obliquity shall be 0 $\pm$ 5 degrees. The angle of obliquity is determined by the angle formed by a perpendicular to a plane tangent to the point of impact and line of flight of the projectile. The obliquity can be determined by any suitable method, however, it is important that the obliquity is determined with respect to the line of flight, and not any other reference. Once the point of impact has been determined on the test sample, and the obliquity fixed, the gun and velocity screen distances must not be changed until the test has been completed.

30.5 Testing.

30.5.1 Test, ballistic. The first round shall be loaded with the reference propellant charge and fired into the sample, and the reading on the chronograph recorded. The velocity shall be computed and recorded. The witness plate shall then be examined for penetration. A complete penetration is recorded when the witness plate is held up to a light and any penetration is noted, regardless of what has penetrated the witness plate. If no light is visible through the witness plate, a partial penetration is recorded. If the first round fired yields a complete penetration, for the second round use a propellant charge equal to that of the first minus the propellant decrement of 50 or 100 ft/sec in an attempt to obtain a partial penetration. If the first round yields a partial penetration, use a propellant charge for the second equal to that of the first plus a propellant increment for 50 or 100 ft/sec in an attempt to obtain a complete penetration. A propellant increment or decrement, as applicable, for at least 50 ft/sec should be used until one partial and one complete penetration are obtained. After obtaining a partial and complete penetration, the propellant increment or decrement for 50 ft/sec should be used. Continue firing using this up (on a partial penetration) and down (on a complete penetration) method until at least 3 complete and 3 partial penetrations having a velocity spread not greater than 125 ft/sec is obtained (see 30.5.2). This up-and-down method of firing must be followed in order to achieve an acceptable ballistic limit (V_{50}). No further testing shall be conducted on the sample unit as soon as this condition has been achieved. A V_{50} protection ballistic limit shall be determined from single round fair impacts on each of at least 6 complete armor inserts of the ballistic sample unit. A fair impact results when an unyawed projectile strikes an unsupported area of the ballistic test plate at least 1-1/2 inches from the outside edge of the ceramic component. (The impacted area shall contain no belts, buckles, webbing, etc.)

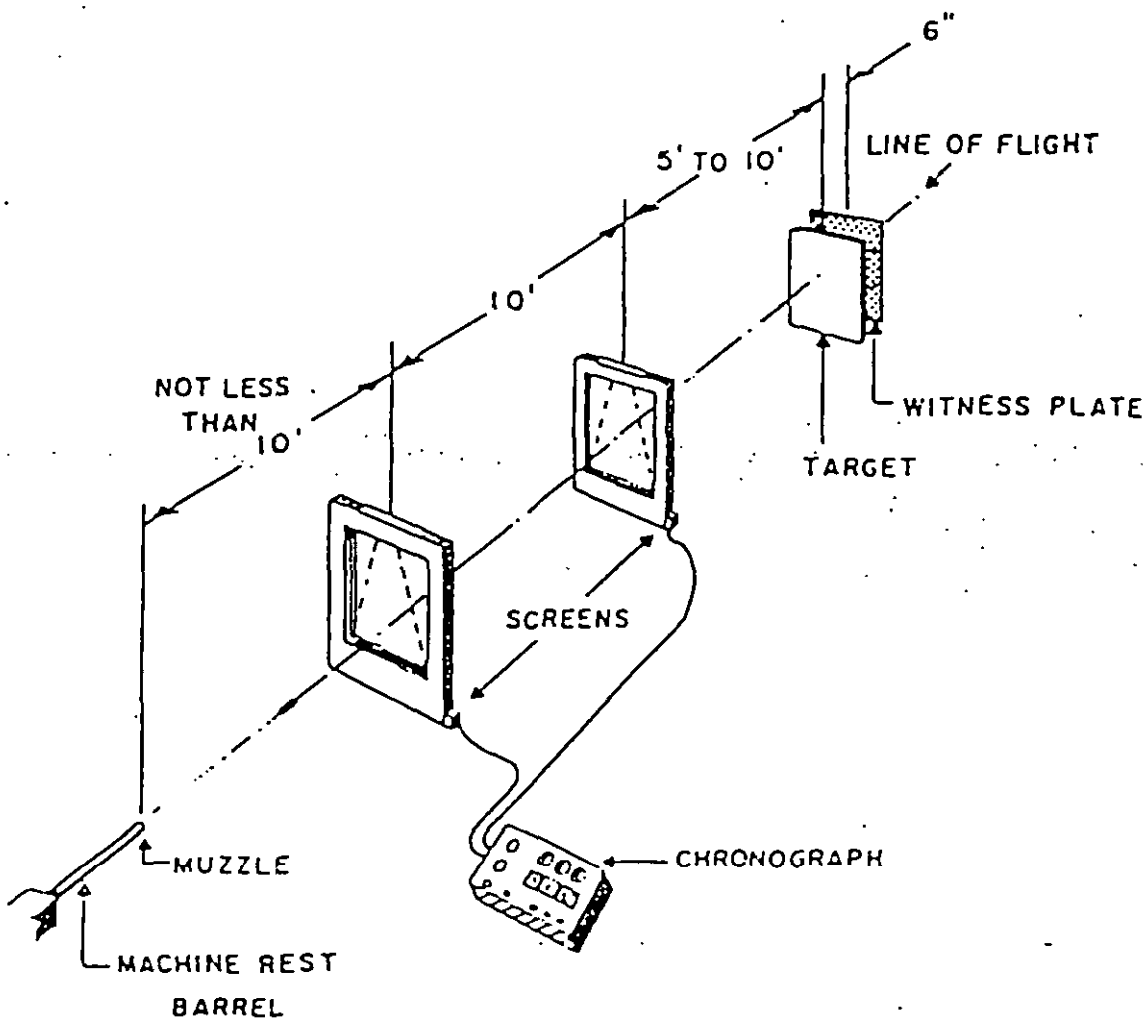
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APPENDIX A

30.5.2 Calculation of ballistic limit. The ballistic limit (V_{50}) is calculated by taking the arithmetic mean of the 3 lowest velocities producing complete penetration and the 3 highest velocities producing partial penetration, provided the velocity spread for the 6 rounds is not greater than 125 ft/sec. If, after all samples in the sample unit have been tested and it was not possible to obtain the 3 lowest velocities for complete penetrations and the 3 highest velocities for partial penetration within a spread of 125 ft/sec, then the ballistic limit (V_{50}) calculation shall be based on one of the following two alternatives. First, the 3 lowest velocities for complete penetration and the 3 highest velocities for partial penetration within a spread of 150 ft/sec; and second, if the first alternative cannot be applied, the 2 lowest velocities for complete penetration and the 2 highest velocities for partial penetration with a 150 ft/sec spread. If none of the preceding ballistic limit (V_{50}) calculation stipulations regarding test velocity spreads apply, then all the valid test velocity values shall be examined to determine whether there are at least 3 partial penetration velocities above 2850 ft/sec and that there are no complete penetration velocities lower than 2850 ft/sec. If these conditions are satisfied, the sample unit will be considered to have met the ballistic limit (V_{50}) requirement. If none of the acceptable calculation procedures can be applied, and if the analysis of the individual test velocities show that they do not meet the requirement for partial penetration velocities higher than 2850 ft/sec and no complete penetration velocities lower than 2850 ft/sec, then the sample unit shall be considered to have failed the ballistic limit (V_{50}) requirement. The ballistic limit (V_{50}) must be at least 2850 ft/sec in order to pass the ballistic test requirement.

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FIGURE 2. BALLISTICS FACILITY LAYOUT



NOTE: MEASUREMENTS TO BE TAKEN TO THE NEAREST 0.01 FOOT

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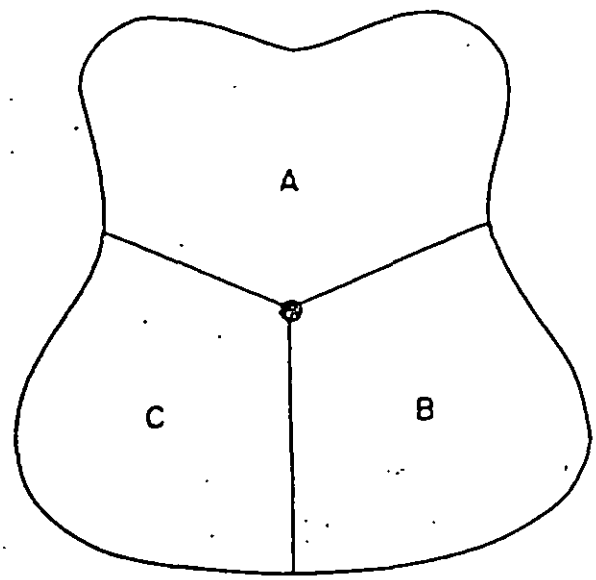


FIGURE 3

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APPENDIX B

Radiographic requirements for examination of ceramic body armor inserts

10. SCOPE

10.1 Scope. This appendix details radiographic inspection process to be used for the examination of ceramic body armor materials. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

SPECIFICATIONS

MILITARY

MIL-R-11470 - Radiographic inspection: Qualification of equipment operators and procedures

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic

30. REQUIREMENTS

30.1 Application. Radiographic inspection is applicable for the detection of internal discontinuities in ceramic body armor inserts and concurrently providing a permanent and ready future reference of internal conditions.

30.2 Equipment. Equipment for radiographic inspection shall consist of the following items:

a. 150 kv X-Ray Equipment.
b. Transmission Densitometer.
c. Penetrameters and Stepwedges.
d. Fine grain film and film envelope.
e. Lead letters and numerals for X-ray film/part identification.
f. Sheet lead 1/8 inch thick to conform to ceramic armor insert.
g. High intensity radiographic viewer.
h. 10X magnifier and film processing facilities.

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APPENDIX B

30.3 Test method. Unless otherwise specified, all radiographic test equipment and procedures shall conform to the requirements of MIL-R-11470 and MIL-STD-453. When other inspection methods are used in addition to X-ray, the X-ray inspection shall be performed prior to more expensive methods and subsequent to less expensive methods. Parts or material showing evidence of inadequate visual or penetrant inspection or parts not in a suitable condition for radiographic inspection shall not be inspected unless properly prepared and recycled.

30.4 Radiograph identification. Each radiograph shall be permanently marked with the name of the test laboratory, X-ray control or test number, part serial number, if any, and view number. All identification markings shall be outside the area being inspected whenever practical. Each radiograph envelope shall contain no more than 12 radiographs and shall list plate serial numbers of the radiographs contained therein. Each envelope shall also bear a rubber stamp image outlining exposure factors and technique employed.

RADIOGRAPHIC TEST RECORD

Date _____ Material _____ Thickness _____

Kilo Voltage _____ M.A _____ Exp. Time _____

Film Type _____ Screens _____

FF Distance _____ Equipment _____

H&D Density _____ No. of Exp. _____

Remarks _____

30.5 Procedures.

30.5.1 Radiographic coverage. Two radiographs, each representing at least half of each ceramic plate, shall be made to effect 100 percent radiographic coverage. In order to avoid image distortion, each exposure shall be made with the ceramic plate angulated or propped so as to be as normal as possible to the X-ray beam.

30.5.2 Radiographic technique. A standard X-ray technique for each ceramic material and type or design shall be established subject to the approval of the procuring agency. All radiographs shall be in the H&D density range of 1.5 -2.2 + 0.20. The radiographic quality level shall be 2-2T. Radiographic equivalence systems shall be permitted using stopwedges and penetrameters exhibiting the same radiographic attenuation properties as ceramic armor materials. The suggested X-ray techniques for high purity boron carbide ceramic armor materials and composites shall be as follows:

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APPENDIX B

Ceramic Plate only - 50 kv, 15 Ma, 40"FFD, 2'0", Kodak M, NS
Complete Composite - 70 kv, 5 Ma, 40"FFD, 1'45", Kodak M, NS

The suggested X-ray techniques for modified boron carbide ceramic armor materials and composites shall be as follows:

Ceramic Plate only - 60 kv, 18 Ma, 40"FFD, 1'50", Kodak M, NS
Complete composite - 70 kv, 15 Ma, 40"FFD, 1'35", Kodak M, NS

The suggested X-ray techniques for silicon carbide ceramic armor composites shall be as follows:

Complete Composite - 80 kv, 10 Ma, 40"FFD, 1'30", Kodak M, NS

The suggested X-ray techniques for aluminum oxide ceramic armor composites shall be as follows:

Complete Composite - 80 kv, 10 Ma, 40"FFD, 1'30", Kodak M, NS

Penetrameters shall be placed so as not to obscure X-ray images. Penetrameter holes and outlines shall be as specified in MIL-STD-453.

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APPENDIX C

Liquid Penetrant Requirements for the Examination of Ceramic Body Armor Inserts

10. SCOPE

10.1 Scope. This appendix details the liquid penetrant inspection process to be used for the examination of ceramic body armor materials. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

SPECIFICATIONS

MILITARY

MIL-I-25135 - Inspection Materials, Penetrant.

20.2 Other publications.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 165 - Liquid penetrant inspection

30. REQUIREMENTS

30.1 Application. Liquid penetrant inspection is applicable for the detection of surface discontinuities and shall be applied prior to radiography.

30.2 Equipment. The equipment required shall be capable of performing uniform controlled operation using ASTM test procedures A1 and B3 per ASTM E165.

30.3 Penetrant materials. For procedure A1 per ASTM E165, Magnaflux ZL-17B penetrant shall be used. For procedure B3 per ASTM E165, Magnaflux SKL-HF penetrant shall be used. Equivalent materials complying with MIL-I-25135 may be used.

30.4 Test method. The test procedures as outlined in ASTM E165 (procedure A1 for silicon carbide, high purity and modified boron carbide and procedure B3 for aluminum oxide) shall be used. All discontinuities detected shall be marked for further evaluation or disposition. A ten power magnifying lens with or without illumination shall be used as necessary to assure accurate detection of indications.

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APPENDIX D

Visual inspection requirements for the examination of ceramic body armor inserts

10. SCOPE

10.1 Scope. This appendix details the visual inspection process to be used for the examination of ceramic body armor materials. This appendix is a mandatory part of the specifications. The information herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Application. Visual inspection is applicable for the detection of surface discontinuities and shall be applied prior to any other nondestructive test.

30.2 Equipment. The equipment for visual inspection shall consist of the following items:

a. Suitable work bench.
b. Ten-power magnifier.
c. Suitable lighting (can be part of magnifier).
d. Marking pens or crayons.
e. Scales or graduated magnifiers for recording diameter of flaw.
f. Suitable gage for measuring depth of flaw.

30.3 Test method. Each ceramic insert shall be carefully examined visually using ten power magnification where necessary. Flaw type indications shall be identified and measured. All discontinuities detected shall be marked for further evaluation or disposition.

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